

HW3: Confidence Intervals and Comparisons of Means

Due: Mon Mar 4, 2024 11:59pm

Answer the following questions. Use the sample code provided in class. You should submit a word document with your answers to each question and any graphs you are asked to generate imported as well as your RScript file. Make sure both files have your name_HW3 as the file name (i.e. SKemper_HW3).

The gender inequality index (GII) provides insights into gender disparities in health, empowerment and the labor market. The GII is a composite measure, reflecting inequality in achievements between women and men in three dimensions: reproductive health, empowerment and the labor market.

- The health dimension is measured by the maternal mortality ratio and the adolescent fertility rate.
- The empowerment dimension is measured by the share of parliamentary seats held by each gender, and by secondary and higher education attainment levels.
- The labor dimension is measured by women's participation in the workforce.

The GII varies between 0 and with higher values representing more equal outcomes. It is designed to reveal the extent to which national human development achievements are eroded by gender inequality, and to provide empirical foundations for policy analysis and advocacy efforts. [Code help](#)

[Download Code help](#)

1. Import the [data table](#)
2. [Download data table](#)
3. and look at it. Do you see any issues you should deal with? (HINT: there are some missing values)
4. Create a histogram and a smoothed density estimate of the 2006, 2010, and 2013 scores.
5. Next create a box and whisker plot of the same data.
6. Calculate summary data for each year you have plotted. Have the means changed over time? How about the standard deviations?
7. Do the data look normally distributed based on what you know thus far?
8. Calculate the z-scores for each of the years (2006, 2010, 2013).
9. Create a confidence interval for the mean for each of the years (2006, 2010, 2013).
10. Re-Import the data and remove the rows with missing data using `na.omit` if you haven't already.

11. What test would you use to compare the global average GII for 2006 to the global average GII for 2013 and why?
12. Set up this test in words and mathematically as shown in class.
13. Run this test. What are your results? Do your confidence intervals for these two years overlap?